# UNIVERSITY OF THE PEOPLE

**CS 1103-01, AY2026-T4**

**Unit 7 Programming Assignment**

**Generic Library Catalog**

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Date: May 14, 2026

## 1. Assignment Overview

This assignment demonstrates the use of generic classes and methods in Java to create a versatile Library Catalog system. The system can manage different types of library items, including books, DVDs, and magazines using a single, reusable generic design. The implementation follows core object-oriented principles and Java Collection Framework best practices.

## 2. Program Design and Structure

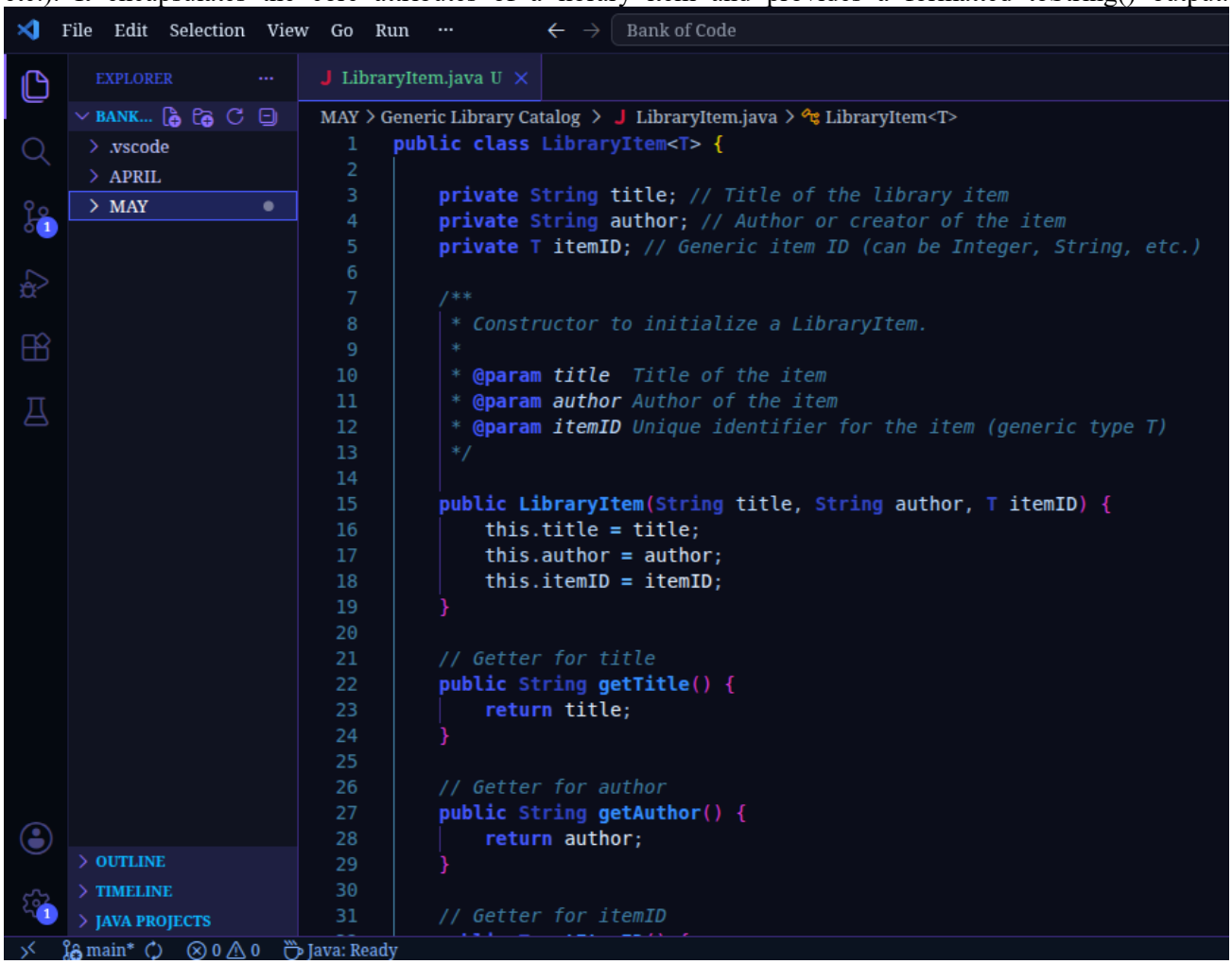
The program consists of three Java classes, each with a distinct responsibility:

| Class | File | Purpose |
|---|---|---|
| LibraryItem<T> | LibraryItem.java | Generic class representing a library item (title, author, itemID) |
| GenericCatalog<T> | GenericCatalog.java | Generic catalog with add, remove, retrieve, and display methods |
| Main | Main.java | Command-line interface for user interaction and testing |

# 3. Java Source Code

## 3.1 LibraryItem.java, Generic Item Class

The LibraryItem<T> class uses a type parameter T to allow flexible item IDs of any object type (Integer, String, etc.). It encapsulates the core attributes of a library item and provides a formatted toString() output.

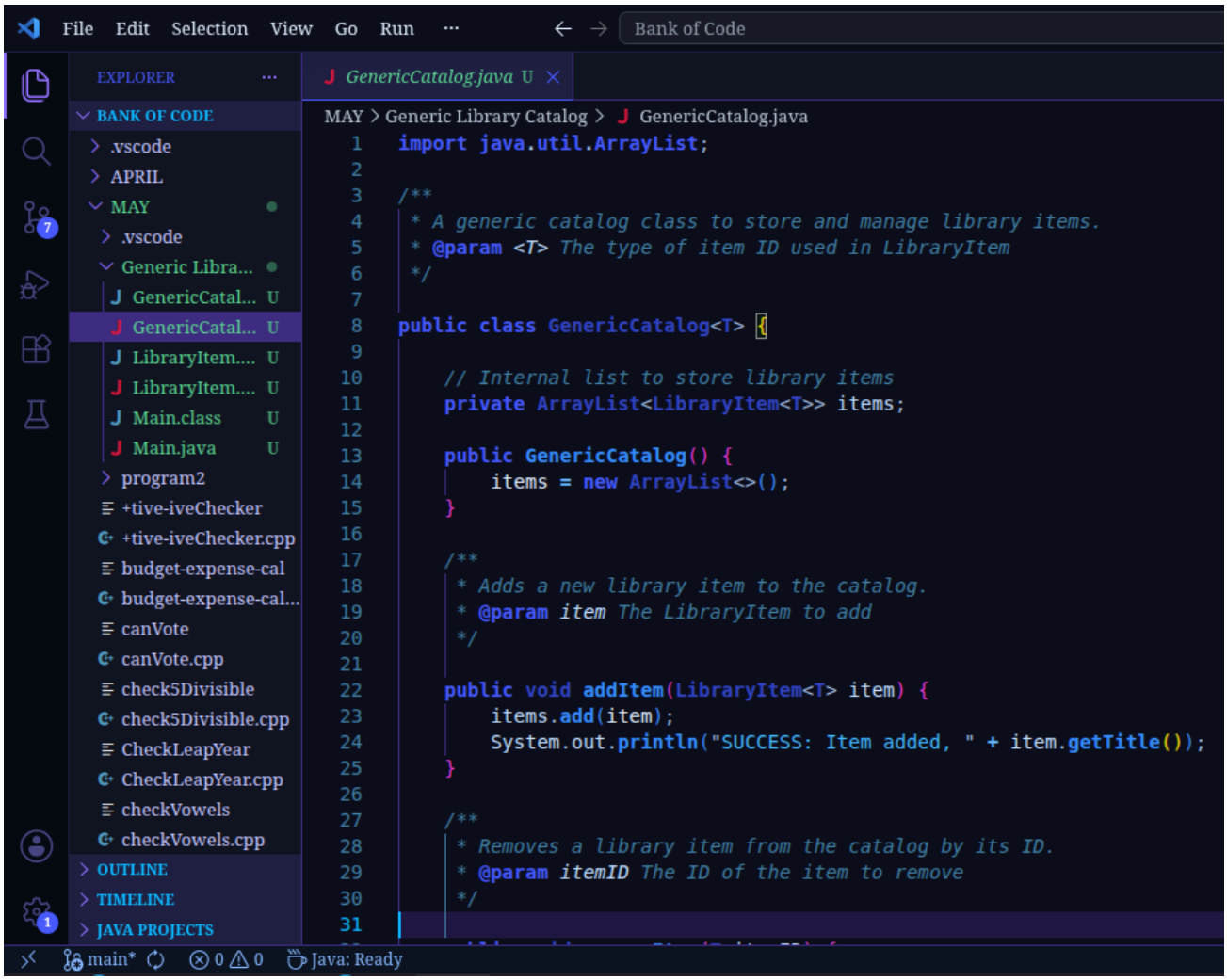



**Source Code:**

https://github.com/mubashir1837/APRIL/blob/main/MAY/Generic%20Library%20Catalog/LibraryItem.java

## 3.2 GenericCatalog.java, Generic Catalog Class

The GenericCatalog<T> class uses an ArrayList internally to store LibraryItem<T> objects. It provides methods to add, remove, retrieve, and display items, along with proper error handling for edge cases such as removing a non-existent item.

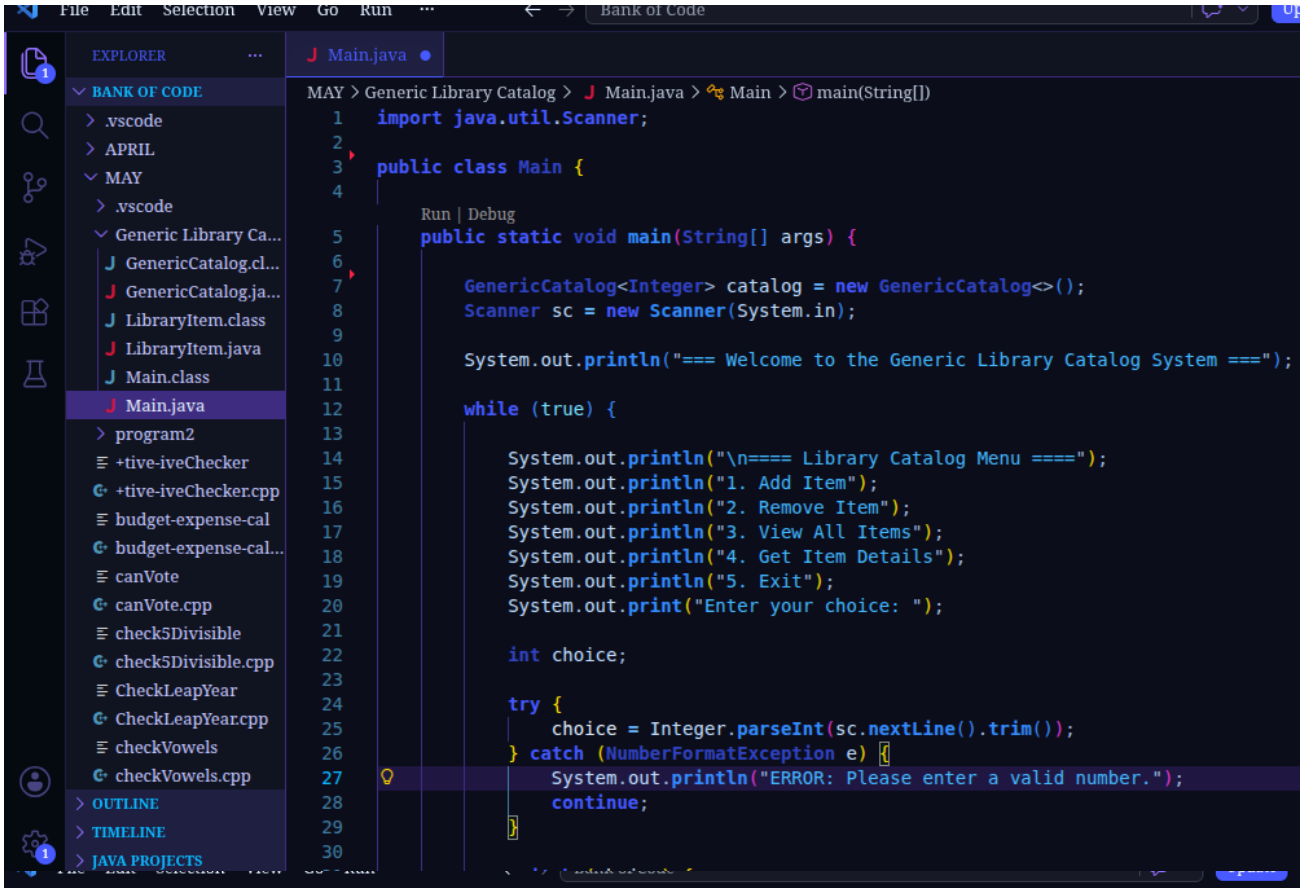


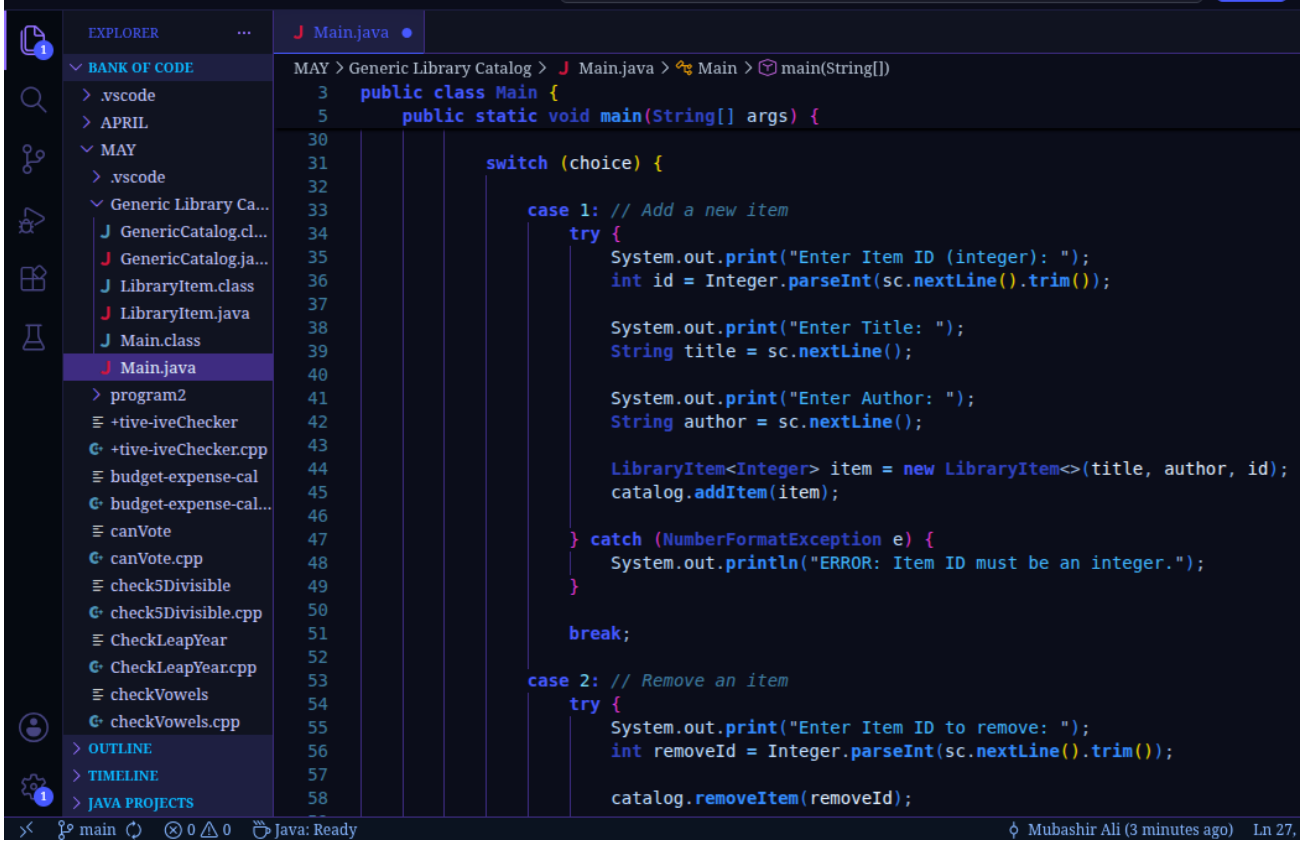

**Source Code:**

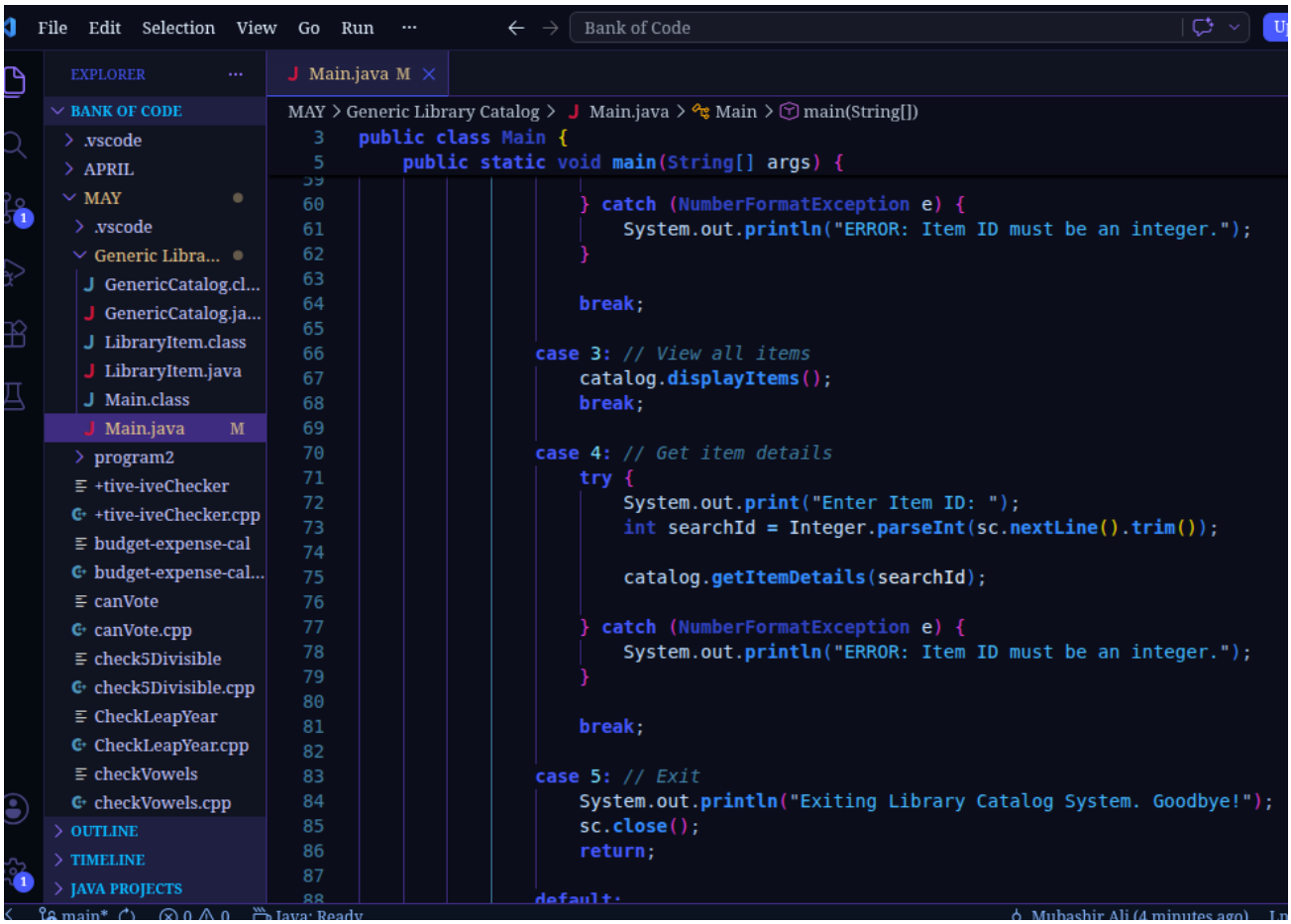
https://github.com/mubashir1837/APRIL/blob/main/MAY/Generic%20Library%20Catalog/GenericCatalog.java

### 3.3 Main.java, Command-Line Interface

The Main class provides a menu-driven command-line interface. It creates a GenericCatalog<Integer> and allows the user to interact with the catalog through numbered menu options. Exception handling is included for invalid input.


File Edit Selection View Go Run ...
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EXPLORER
Main.java
BANK OF CODE
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APRIL
MAY
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GenericCatalog.ja...
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LibraryItem.java
Main.class
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canVote
canVote.cpp
check5Divisible
check5Divisible.cpp
CheckLeapYear
CheckLeapYear.cpp
checkVowels
checkVowels.cpp
OUTLINE
TIMELINE
JAVA PROJECTS
MAY > Generic Library Catalog > Main.java > Main > main(String[])
1 import java.util.Scanner;
2
3 public class Main {
4
Run | Debug
5 public static void main(String[] args) {
6
7 GenericCatalog<Integer> catalog = new GenericCatalog<>();
8 Scanner sc = new Scanner(System.in);
9
10 System.out.println("=== Welcome to the Generic Library Catalog System ===");
11
12 while (true) {
13
14 System.out.println("\n==== Library Catalog Menu ====");
15 System.out.println("1. Add Item");
16 System.out.println("2. Remove Item");
17 System.out.println("3. View All Items");
18 System.out.println("4. Get Item Details");
19 System.out.println("5. Exit");
20 System.out.print("Enter your choice: ");
21
22 int choice;
23
24 try {
25 choice = Integer.parseInt(sc.nextLine().trim());
26 } catch (NumberFormatException e) {
27 System.out.println("ERROR: Please enter a valid number.");
28 continue;
29 }
30


EXPLORER
Main.java
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.vscode
APRIL
MAY
.vscode
Generic Library Ca...
GenericCatalog.cl...
GenericCatalog.ja...
LibraryItem.class
LibraryItem.java
Main.class
Main.java
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+tive-iveChecker.cpp
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budget-expense-cal...
canVote
canVote.cpp
check5Divisible
check5Divisible.cpp
CheckLeapYear
CheckLeapYear.cpp
checkVowels
checkVowels.cpp
OUTLINE
TIMELINE
JAVA PROJECTS
MAY > Generic Library Catalog > Main.java > Main > main(String[])
3 public class Main {
5 public static void main(String[] args) {
30
31 switch (choice) {
32
33 case 1: // Add a new item
34 try {
35 System.out.print("Enter Item ID (integer): ");
36 int id = Integer.parseInt(sc.nextLine().trim());
37
38 System.out.print("Enter Title: ");
39 String title = sc.nextLine();
40
41 System.out.print("Enter Author: ");
42 String author = sc.nextLine();
43
44 LibraryItem<Integer> item = new LibraryItem<>(title, author, id);
45 catalog.addItem(item);
46
47 } catch (NumberFormatException e) {
48 System.out.println("ERROR: Item ID must be an integer.");
49 }
50
51 break;
52
53 case 2: // Remove an item
54 try {
55 System.out.print("Enter Item ID to remove: ");
56 int removeId = Integer.parseInt(sc.nextLine().trim());
57
58 catalog.removeItem(removeId);
main 0 0 Java: Ready Mubashir Ali (3 minutes ago) Ln 27,




**Source Code:**

https://github.com/mubashir1837/APRIL/blob/main/MAY/Generic%20Library%20Catalog/Main.java

## 4. How Generics Are Applied

This program demonstrates generics in two key ways:

- Type Parameter T in LibraryItem<T>: The itemID field uses type T, allowing the same class to store Integer IDs, String IDs, or any other object type without rewriting the class.
- Type Parameter T in GenericCatalog<T>: The catalog stores a list of LibraryItem<T> objects, ensuring that the catalog and its items always use the same ID type. This enforces type safety at compile time.
- Code Reusability: A single GenericCatalog class works for books with Integer IDs, magazines with String IDs, DVDs with Long IDs, and any future item type, with zero code duplication.
- Compile-Time Safety: Unlike raw ArrayList (pre-generics), the compiler prevents mixing incompatible item types in the same catalog.

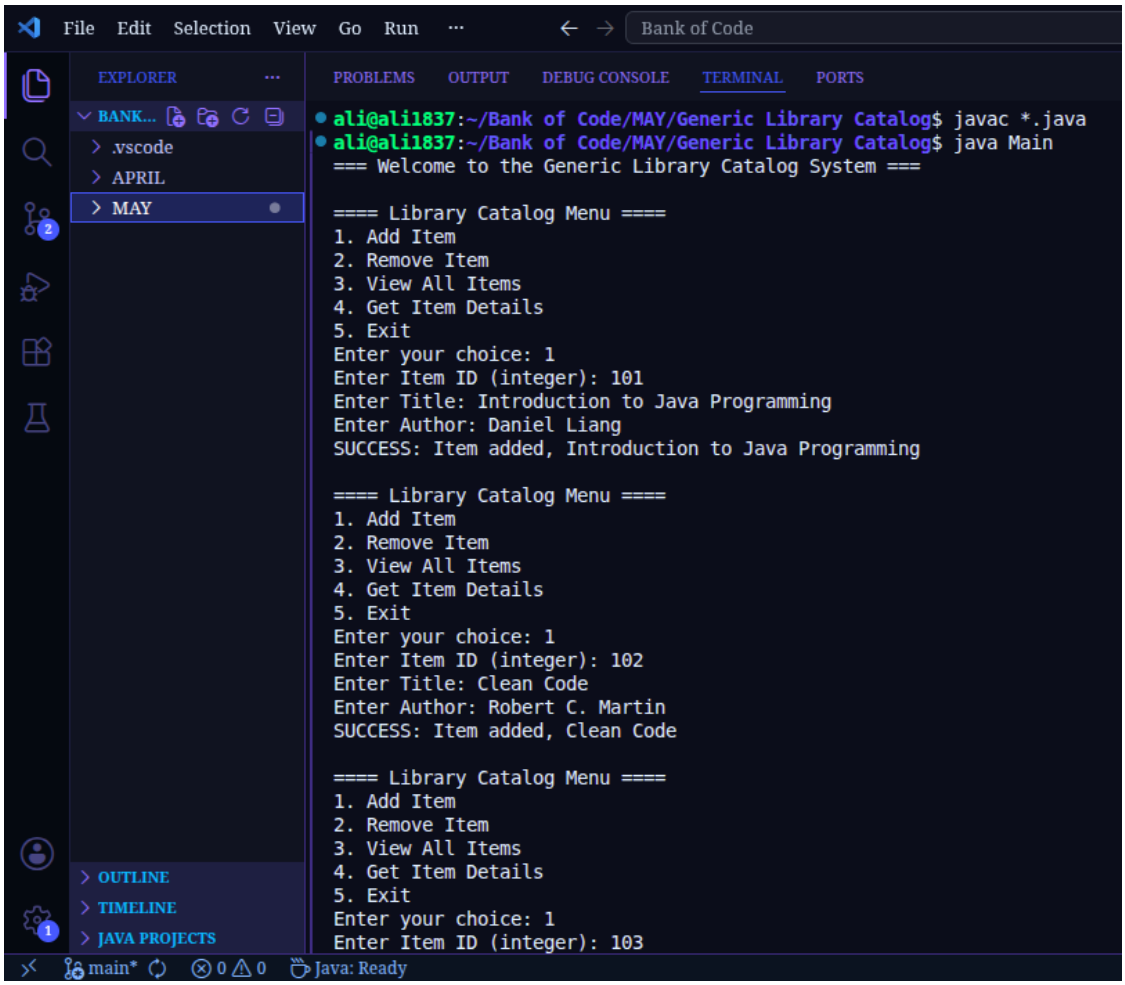
## 5. Error Handling

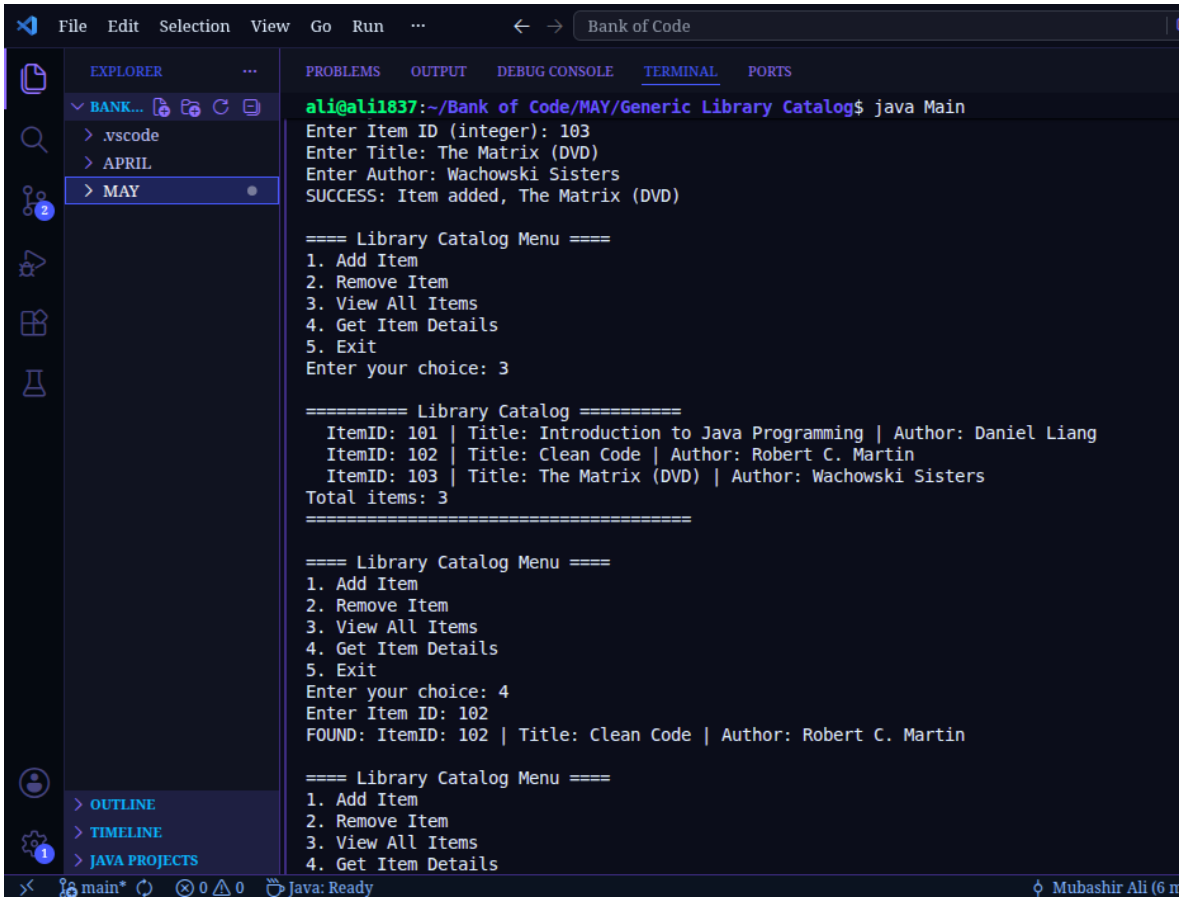
The program implements error handling in the following scenarios:

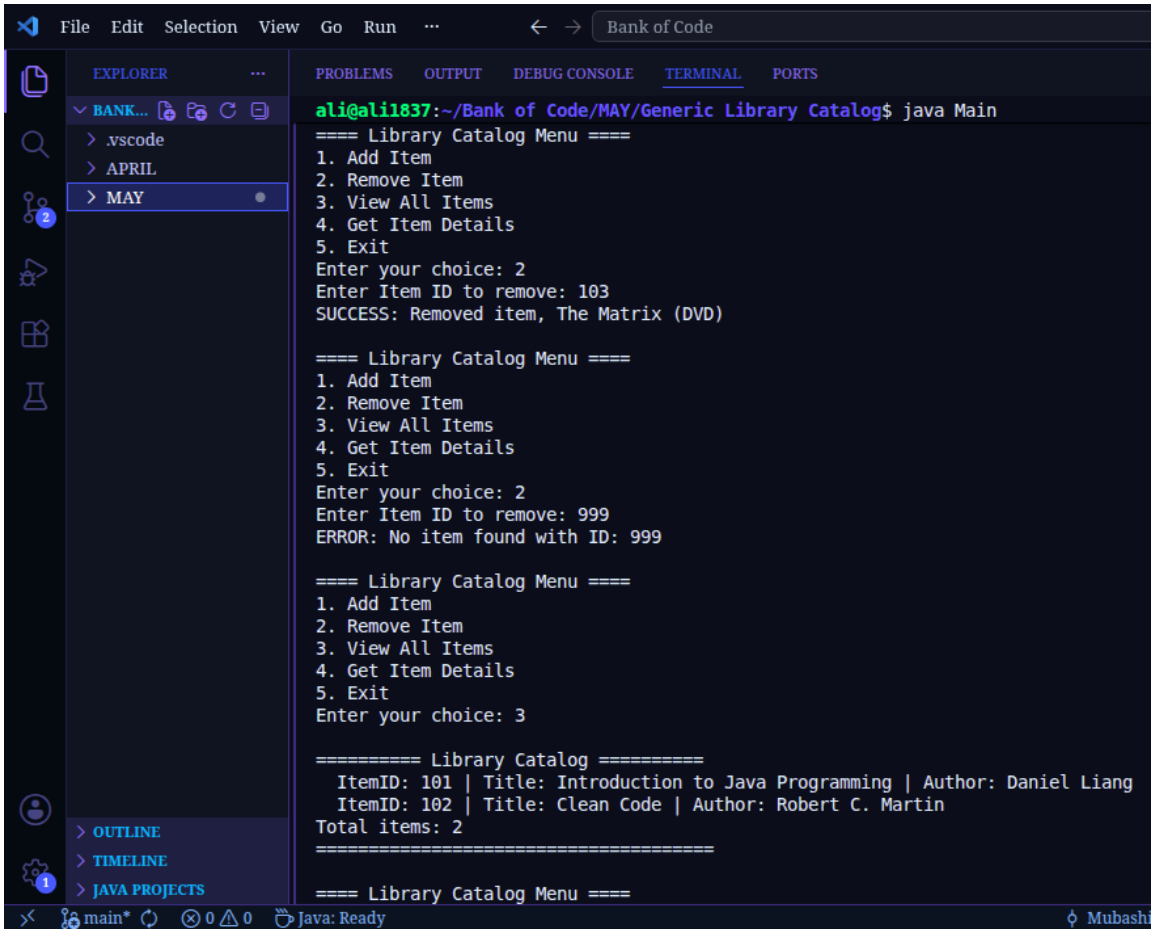
- Removing a non-existent item: If the specified item ID is not found in the catalog, the removeItem() method prints a clear error message: "ERROR: No item found with ID: [id]" and continues without crashing.
- Invalid menu input: The main menu uses a try-catch block around Integer.parseInt() to gracefully handle non-numeric input, printing an error message and returning to the menu.
- Empty catalog display: If displayItems() is called on an empty catalog, it prints "INFO: The catalog is currently empty." instead of showing a blank list.

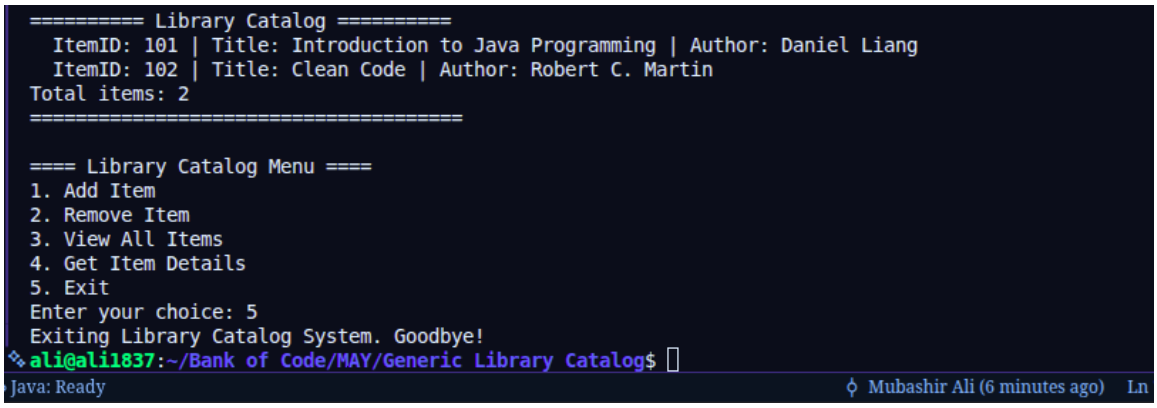
## 6. Program Output

The following output demonstrates all operations, adding three items (2 books, 1 DVD), viewing the catalog, retrieving details, removing an item, attempting to remove a non-existent item, and viewing the updated catalog:




File Edit Selection View Go Run ···
Bank of Code
EXPLORER
BANK...
.vscode
APRIL
MAY
OUTLINE
TIMELINE
JAVA PROJECTS
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
ali@ali1837:~/Bank of Code/MAY/Generic Library Catalog$ java Main
Enter Item ID (integer): 103
Enter Title: The Matrix (DVD)
Enter Author: Wachowski Sisters
SUCCESS: Item added, The Matrix (DVD)
==== Library Catalog Menu ====
1. Add Item
2. Remove Item
3. View All Items
4. Get Item Details
5. Exit
Enter your choice: 3
========== Library Catalog ==========
ItemID: 101 | Title: Introduction to Java Programming | Author: Daniel Liang
ItemID: 102 | Title: Clean Code | Author: Robert C. Martin
ItemID: 103 | Title: The Matrix (DVD) | Author: Wachowski Sisters
Total items: 3
==== Library Catalog Menu ====
1. Add Item
2. Remove Item
3. View All Items
4. Get Item Details
5. Exit
Enter your choice: 4
Enter Item ID: 102
FOUND: ItemID: 102 | Title: Clean Code | Author: Robert C. Martin
==== Library Catalog Menu ====
1. Add Item
2. Remove Item
3. View All Items
4. Get Item Details
main*
0 0
Java: Ready
Mubashir Ali (6 m




## 9. References

Eck, D. J. (2022). *Introduction to programming using Java version 9, JavaFX edition*. Licensed under CC 4.0. https://math.hws.edu/javanotes/

Horstmann, C. S. (2023). *Core Java Volume I: Fundamentals* (12th ed.). Pearson.

Oracle. (2024). Java Generics (The Java Tutorials). https://docs.oracle.com/javase/tutorial/java/generics/

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